CAREERS 360

PREPARATION Series

NEET UG 2027

Biotechnology and Its Applications PYQs

100+ Previous Year Questions with Detailed Solutions

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Biotechnology and Its Applications Previous Year Questions

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About this eBook

Dear NEET Aspirants,

Welcome to your ultimate Year-wise NEET PYQs Guide for Biotechnology and Its Applications! This eBook is designed to help you master one of the critical sections of Biology, which holds significant importance in the NEET exam. By focusing on previous years' questions (PYQs) organized year-wise, this eBook ensures that you understand the exam pattern, frequently tested topics, and question trends over the years.

What does this eBook contain?

Year-wise NEET PYQs Compilation – We have systematically arranged previous years' questions year-wise to help you analyze the trends and understand how concepts are tested over different NEET exams.

Detailed Solutions – Every question comes with a step-by-step explanation to ensure conceptual clarity, reinforcing the fundamentals of Biotechnology and Its Applications.

Topic-Wise Yearly Breakdown – The eBook covers the following topics:

- **Principles of Biotechnology** – Year-wise PYQs on recombinant DNA technology, gene cloning, and gene expression.
- **Applications of Biotechnology** – NEET questions on genetic engineering, biotechnology in agriculture, medicine, and environmental conservation.
- **Genetic Engineering** – Questions from past years on gene editing tools like CRISPR/Cas9 and their applications.
- **Biotechnology in Medicine** – Yearly NEET PYQs on vaccines, diagnostics, and gene therapy.
- **Ethical Issues in Biotechnology** – Questions covering biosafety, bioethics, and intellectual property rights.

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How Will This eBook Benefit You?

- **NEET-Oriented Preparation** – By focusing on **PYQs**, you'll develop a deep understanding of how questions are framed and what to expect in the exam, particularly in the biotechnology section.
- **Higher Accuracy & Speed** – Practicing past questions improves your **problem-solving speed** and accuracy, giving you an edge over others in tackling complex biotechnology concepts.
- **Boosts Confidence** – Since these questions have appeared in **previous NEET exams**, solving them will strengthen your confidence and enhance your exam strategy, especially in areas like genetic engineering and biotechnology applications.
- **Smart & Efficient Learning** – Instead of studying randomly, this eBook ensures structured, high-yield learning, helping you focus on scoring topics like recombinant DNA technology and biotechnology applications in medicine and agriculture without unnecessary distractions.

Mastering **Biotechnology and Its Applications PYQs** will give you the competitive advantage needed to excel in NEET, helping you stay consistent and take one step closer to your dream of becoming a medical professional!

Importance Of Biotechnology and Its Applications In NEET

The unit Biotechnology and Its Applications is crucial for NEET aspirants as it covers key concepts like genetic engineering and recombinant DNA technology, which are essential for understanding how living organisms are used to create innovative products and processes. This unit is divided into two main chapters: Principles of Biotechnology and Applications of Biotechnology. The first chapter focuses on the fundamental processes and techniques used in biotechnology, such as gene manipulation and DNA recombination. The second chapter explores how biotechnology is applied in various fields like medicine (e.g., vaccines and therapeutics), agriculture (e.g., genetically modified crops), and industry (e.g., biofuels and bioremediation). Mastering these concepts is vital for scoring well in NEET, as they form the basis for careers in biotechnology research and related fields.

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Weightage Of Biotechnology And Its Applications In NEET

Biotechnology holds considerable weightage in NEET and comprises roughly 7.5% of the marks in the Biology section. Three questions usually come from this chapter and carry 12 marks together. This much weightage makes this chapter a "do-or-die" for students who dream of scoring high marks in NEET. The fact that questions from this chapter have appeared in previous years is an example of how crucial this chapter is for the exam.

NEET 2026

Q1. The human protein named α-1-antitrypsin, obtained from transgenic animals, is used for the treatment of _____

(A) Alzheimer's disease
(B) Rheumatoid arthritis
(C) Emphysema
(D) Cystic fibrosis

Correct Answer: (C) Emphysema

Explanation: α-1-antitrypsin deficiency lets unchecked protease enzymes damage lung tissue, causing emphysema. Transgenic animals are engineered to produce this human protein, which is then used therapeutically to protect lung tissue in emphysema patients.

Q2. The toxin proteins isolated from *Bacillus thuringiensis*, coded by which of the following genes would control cotton bollworms and corn borers, respectively?

(A) cryIAc and cryIIb
(B) cryIAc and cryIIIAb
(C) cryIAc and cryIAb
(D) cryIIAb and cryIAc

Correct Answer: (C) cryIAc and cryIAb

Explanation: *Bacillus thuringiensis* produces different Cry toxin proteins encoded by

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different cry genes. The cryIAc gene product targets and controls cotton bollworms, while the cryIAb gene product controls corn borers — the basis of Bt cotton and Bt corn pest resistance.

RE NEET 2026

Q1. The inactive form of Bt toxin is converted to the active form in the insect gut _____.

(A) Due to acidic pH
(B) By proteases
(C) By nucleases
(D) Due to alkaline pH

Correct Answer: (D) Due to alkaline pH

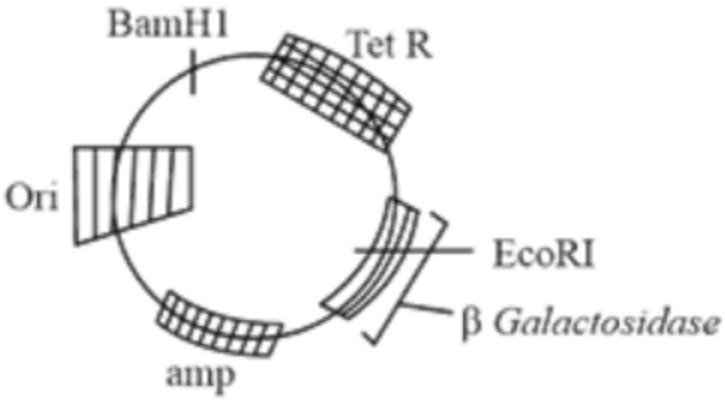
Explanation: Bt toxin is produced by *Bacillus thuringiensis* as an inactive protoxin. Inside the insect gut, which has an alkaline pH, the protoxin is solubilised and then proteolytically cleaved into its active, insecticidal form that damages the gut epithelium.

NEET 2025

Chapter: Biotechnology and Its Applications

1. In the above-represented plasmid alien piece of DNA is inserted at the E. coli site. Which of the following strategies will be chosen to select the recombinant colonies?

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Option 1. Blue colour colonies grown on ampicillin plates can be selected.

Option 2. Using ampicillin and tetracycline containing medium plate.

Option 3. Blue colour colonies will be selected.

Option 4. White colour colonies will be selected.

Difficulty Level: Medium

Answer: In plasmids such as pUC19, the lacZ gene is employed in blue-white screening, a technique for the selection of recombinant colonies. If foreign DNA is introduced into the lacZ region, the gene is disrupted, and β-galactosidase production stops.

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Thus, recombinant plasmid-containing colonies are white because X-gal is not able to be cleaved. Non-recombinant colonies, however, have a lacZ gene, which produces β-galactosidase, and are blue. Hence, white colonies are chosen during screening as they most probably have the inserted foreign DNA.

Hence, the correct answer is option 4) White colour colonies will be selected.

2. Which of the following genetically engineered organisms was used by Eli Lilly to prepare human insulin?

Option 1. Phage

Option 2. Bacterium

Option 3. Yeast

Option 4. Virus

Difficulty Level: Medium
Answer: Eli Lilly is a US pharmaceutical company that manufactures human insulin with recombinant DNA technology as genetically engineered human insulin. Here, genes for human insulin A and B chains were inserted into plasmids separately, which were then introduced into bacteria (Escherichia coli). The genetically changed bacteria produced the insulin chains independently. Later on, the A and B chains were separated, purified, and then chemically merged to create biologically active human insulin.

Hence, the correct answer is option 2) Bacterium.

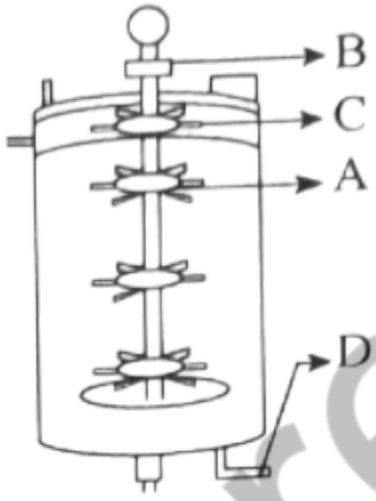
3. Silencing of specific mRNA is possible via RNAi because of
Option 1. Non-complementary ssRNA
Option 2. Complementary dsRNA
Option 3. Inhibitory ssRNA
Option 4. Complementary tRNA
Difficulty Level: Medium

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Answer: RNA interference (RNAi) involves the introduction of double-stranded RNA (dsRNA) that is complementary to the target mRNA. This process gets initiated by the silencing of the target gene through degradation of its mRNA or translational inhibition. The complementary dsRNA is then recognised by the RNA-induced silencing complex (RISC), which results in the degradation of the target mRNA. The process is used as a gene regulatory tool as well as to understand the functions of genes in different research and therapeutic applications.

Hence, the correct answer is option 2)Complementary dsRNA.

4. Identify the part of a bioreactor that is used as a foam breaker from the given figure.



Option 1. C

Option 2. A

Option 3. B

Option 4. D

Difficulty Level: Medium

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Answer: The foam breaker in a stirred-tank bioreactor is located at the top and is labeled as B in the diagram. Foam forms during fermentation and needs to be broken to avoid overflow and maintain aeration efficiency. The foam breaker helps in breaking this foam mechanically. But if excess foam is not controlled, then it can interfere with sensors and valves, which leads to risks and operational failures. Foam beakers allow smooth functioning of the bioreactor.

Hence, the correct answer is Option 3) B.

5. Which of the following enzyme(s) are NOT essential for gene cloning?

A. Restriction enzymes

B. DNA ligase

C. DNA mutase

D. DNA recombinase

E. DNA polymerase

Choose the correct answer from the options given below:

Option 1. B and C only

Option 2. C and D only

Option 3. A and B only

Option 4. D and E only

Difficulty Level: Medium

Answer: A. Restriction enzymes: These are valuable tools of gene cloning. They work like molecular scissors and cleave DNA at a defined sequence to form fragments that are introduced into vectors.

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B. DNA ligase: This is also an important component in gene cloning. It connects the DNA fragment with vector DNA by making the sugar-phosphate backbone, and thus produces a stable recombinant DNA molecule.

C. DNA mutase: These are not needed in gene cloning. DNA mutases are the enzymes that make mutations in DNA, which is not a part of the gene cloning process.

D. DNA recombinase: This enzyme helps in the process of recombination between strands of DNA, which is used in advanced genetic engineering techniques but is not needed in gene cloning.

E. DNA polymerase: Not always, but this tool is often used in PCR to increase the amount of DNA before cloning.

DNA mutase and DNA recombinase are not required in the process of gene cloning.

Hence, the correct answer is option (2) C and D only.

Chapter: Biotechnology: Principles and Processes

1. Which of the following genetically engineered organisms was used by Eli Lilly to prepare human insulin?

Option 1: Phage

Option 2: Bacterium

Option 3: Yeast

Option 4: Virus

Difficulty Level: Medium

Answer: Eli Lilly is a US pharmaceutical company that manufactures human insulin with recombinant DNA technology as genetically engineered human insulin. Here,

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genes for human insulin A and B chains were inserted into plasmids separately, which were then introduced into bacteria (Escherichia coli). The genetically changed bacteria produced the insulin chains independently. Later on, the A and B chains were separated, purified, and then chemically merged to create biologically active human insulin.

Hence, the correct answer is option 2) Bacterium.

2. Polymerase chain reaction (PCR) amplifies DNA following the equation

Option 1. $2N^2$

Option 2. N^2

Option 3. 2^n

Option 4. 2n + 1

Difficulty Level: Medium

Answer: The polymerase chain reaction (PCR) is used to amplify DNA. After each cycle of PCR, the amount of DNA doubles. So, if there are n cycles, the DNA is amplified by a factor of 2^n.

Example - If 4 PCR cycles are run, then 2^4= 2 x 2 x 2x 2=16 DNA fragments will be formed.

Hence, the correct answer is option 3) 2^n

3. Given below are two statements:

Statement I: The DNA fragments extracted from gel electrophoresis can be used in the construction of recombinant DNA.

Statement II: Smaller-sized DNA fragments are observed near the anode, while larger fragments are found near the wells in an agarose gel.

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In light of the above statements, choose the most appropriate answer from the options given below :

Option 1: Statement I is incorrect, but Statement II is correct
Option 2: Both statements I and II are correct
Option 3: Both statements I and II are incorrect
Option 4: Statement I is correct, but Statement II is incorrect
Difficulty Level: Medium
Answer: Statement I is correct: The gel-separated fragments of DNA may be used to construct recombinant DNA. After the fragments have been separated by gel electrophoresis, they may be isolated, purified, and inserted into vectors to form recombinant DNA molecules.

Statement II is correct: Smaller fragments of DNA travel faster and therefore migrate farther in the direction of the anode (positive). On the other hand, larger fragments of DNA travel slowly and are closer to the wells (negative) since they encounter greater resistance when travelling through the gel.

Hence, the correct answer is option (2) Both statements I and II are correct.

4. The blue and white selectable markers have been developed, which differentiate recombinant colonies from non-recombinant colonies based on their ability to produce colour in the presence of a chromogenic substrate.

Given below are two statements about this method:
Statement I: The blue coloured colonies have DNA insert in the plasmid and they are identified as recombinant colonies.
Statement II: The colonies without blue colour have a DNA insert in the plasmid and are identified as recombinant colonies.

In light of the above statements, choose the most appropriate answer from the options given below:

Difficulty Level: Medium

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Option 1: Statement I is incorrect, but Statement II is correct.

Option 2: Both Statement I and Statement II are correct.

Option 3: Both Statement I and Statement II are incorrect.

Option 4: Statement I is correct, but Statement II is incorrect.

Answer: In blue-white screening, a chromogenic substrate such as X-gal is employed. When the lacZ gene is unbroken (no insert of foreign DNA), the enzyme β-galactosidase is formed, and the colonies become blue, non-recombinant. When foreign DNA is added to the lacZ gene, it is broken, and the enzyme is not formed. The colonies are white, a sign of recombinant bacteria. Thus, **Statement I** is false: Blue colonies are not recombinant.

Statement II is true: White colonies are recombinant because they have the DNA insert. This technique can easily identify recombinant and non-recombinant colonies. It is applied extensively in cloning experiments on plasmid vectors. The absence or presence of blue colour is a marker for successful gene insertion.

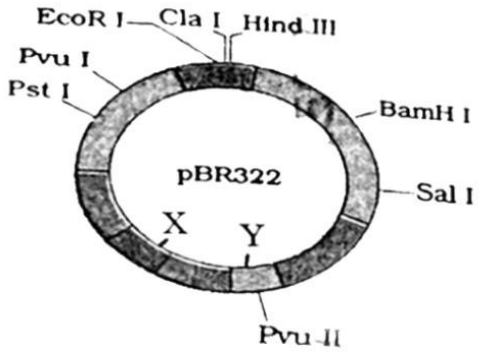
Hence, the correct answer is Option 1) Statement I is incorrect, but Statement II is correct.

NEET 2024

Chapter: Biotechnology: Principles & Processes

1. The following diagram shows restriction sites in E. coli cloning vector pBR322. Find the role of 'X' and 'Y genes:

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Option 1. The gene 'X' is responsible for resistance to antibiotics, and 'Y' is responsible for the protein involved in the replication of plasmid.

Option 2. The gene 'X' is responsible for controlling the copy number of the linked DNA, and 'Y' is for the protein involved in the replication of the plasmid.

Option 3. The gene 'X' is for a protein involved in the replication of plasmid, and 'Yis ' is for resistance to antibiotics.

Option 4. Gene 'X' is responsible for rerecognitiontes, and 'Y' is responsible for antibiotic resistance.

Difficulty Level: Medium

Answer: In the E. coli cloning vector pBR322, the **amp^R gene** (X) provides resistance to ampicillin, allowing the selection of bacterial cells that have successfully taken up the plasmid, while the **tet^R gene** (Y) provides resistance to tetracycline, serving as a marker to differentiate recombinant plasmids from non-recombinant ones when foreign DNA is inserted into specific restriction sites. These selectable markers are essential for identifying and isolating transformants in genetic engineering experiments.

Hence, the correct answer is option 2. The gene 'X' is responsible for controlling the copy number of the linked DNA, and 'Y' is for the protein involved in the replication of Plasmid.

2. Match List I with List II:

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List I	List II
A. α -1 antitrypsin	I. Cotton bollworm
B. Cry IAb	II. ADA deficiency
C. Cry IAc	III. Emphysema
D Enzyme replacement therapy	IV Corn borer

Choose the correct answer from the options given below:

Option 11:A-II, B-I, C-IV, D-III

Option 2: A-III, B-I, C-II, D-IV

Option 3. A-III, B-IV, C-I, D-II

Option 4: A-II, B-IV, C-I, D-III

Difficulty Level: Easy

Answer: A. Emphysema is associated with α-1 antitrypsin (III). Lack of this protein, which protects the lungs, can result in lung conditions, including emphysema.
B. Corn borer (IV) is controlled with Cry IAb. Corn that has undergone genetic modification uses this protein to ward off pests like the corn borer.
C. Cry IAc aids in the battle against cotton bollworm (I). Cotton plants require this protein to defend themselves against cotton bollworms.
D. ADA deficiency is treated with enzyme replacement therapy (II). It aids in the treatment of a disorder in which the body is deficient in an enzyme that causes issues with the immune system.

Hence, the correct answer is option 3) A-III, B-IV, C-I, D-II.

3. What is the fate of a piece of DNA carrying only a gene of interest that is transferred into an alien organism?

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A. The piece of DNA would be able to multiply itself independently in the progeny cells of the organism.

B. It may get integrated into the recipient's genome.

C. It may multiply and be inherited along with the host DNA.

D. The alien piece of DNA is not an integral part of a chromosome.

E. It shows the ability to replicate.

Choose the correct answer from the options given below:

Option 1: A and B only

Option 2. D and E only

Option 3. B and C only

Option 4. A and E only

Difficulty Level: Medium

Answer: The fate of a piece of DNA carrying only a gene of interest which is transferred into an alien organism is:
(B) It may get integrated into the genome of the recipient
(C) It may multiply and be inherited along with the host DNA

This piece of DNA would not be able to multiply itself in the progeny cells of the organism. Still, when it gets integrated into the genome of the recipient, it may multiply and be inherited along with the host DNA.

Hence, the correct answer is option 3) B and C only.

4. Given below are two statements:

Statement I: Bt toxins are insect group-specific and coded by a gene, Cry IAc.

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Statement II: Bt toxin exists as an inactive protoxin in B. thuringiensis. However, after ingestion by the insect, the inactive protoxin gets converted into an active form due to the acidic pH of the insect gut.

In light of the above statements, choose the correct answer from the options given below:

Option 1: Both Statement I and Statement II are true

Option 2. Both Statement I and Statement II are false

Option 3. Statement I is true,e but Statement II is false

Option 4. Statement I is false, but Statement II is true

Difficulty Level: Medium

Answer: The correct answer is option (,3) as specific Bt toxin genes were isolated from Bacillus thuringiensis and incorporated into several crop plants such as cotton. The choice of genes depends upon the crop and the targeted pest, as most Bt toxins are insect-group-specific. The toxin is coded by a gene named cry. There are a number of them; for example, the proteins encoded by the genes cry IAc and cry IIAb control the cotton bollworm, and the protein encoded by cry IAb controls the corn borer.

Hence, the correct answer is Option 3. Statement I is true, but Statement II is false.

NEET 2023

Chapter: Biotechnology and Its Applications

5. Which one of the following techniques does not serve the purpose of early diagnosis of a disease for its early treatment?

Option 1: Recombinant DNA Technology

Option 2: Serum and Urine analysis

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Option 3. Polymerase Chain Reaction (PCR) technique

Option 4. Enzyme-Linked Immuno-Sorbent Assay (ELISA) technique

Difficulty Level: Medium

Answer: Serum and urine analysis are useful diagnostic techniques, but they do not specifically serve the purpose of early diagnosis of a disease for its early treatment. Serum analysis involves testing the blood serum for various markers, such as antibodies or specific substances, to detect the presence of a disease or assess its progression. Urinalysis involves examining urine for abnormalities that may indicate certain diseases or conditions.

Hence, the correct answer is Option 2) Serum and Urine analysis.

6. Transposons can be used during which one of the following?

Option 1: Gene silencing

Option 2. Autoradiography

Option 3. Gene sequencing

Option 4. Polymerase Chain Reaction

Difficulty Level: Difficult

Answer: **Option 1** is correct because the source of complementary RNA for RNAi (RNA interference) or gene silencing could be transposons (mobile genetic elements).

Option 2 is incorrect, as autoradiography usually follows hybridisation

Option 3 is incorrect, as transposons are not required in gene sequencing.

Option 4 is also incorrect, as PCR (polymerase chain reaction) is a process to make copies of DNA and does not need transposons.

Hence, the correct answer is option (1) Gene silencing

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NEET 2022

Chapter: Biotechnology—Principles and Processes

7. During the purification process for recombinant DNA technology, the addition of chilled ethanol precipitates out.

Option 1: RNA

Option 2. DNA

Option 3. Histones

Option 4. Polysaccharides

Difficulty Level: Medium

Answer: During the purification process in recombinant DNA technology, chilled ethanol is often used to precipitate out DNA. This technique is based on the principle that DNA is less soluble in cold ethanol than in aqueous solutions.

When chilled ethanol is added to the DNA solution, it causes the DNA molecules to come out of the solution and form a visible precipitate. This is because the ethanol disrupts the hydrogen bonding and hydrophobic interactions that hold the DNA molecules together, leading to their aggregation and precipitation.

By centrifuging the mixture, the precipitated DNA can be collected as a pellet while other contaminants remain in the supernatant. The DNA pellet can then be further purified and processed for various applications in recombinant DNA technology.

Therefore, Option 2) DNA is the correct answer for the component that precipitates out when chilled ethanol is added during the purification process for recombinant DNA technology.

Hence, the correct answer is option 2) DNA.

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8. Upon exposure to UV radiation, DNA stained with ethidium bromide will show

Option 1: Bright red colour

Option 2: Bright blue colour

Option 3. Bright yellow colour

Option 4. Bright orange colour

Difficulty Level: Medium

Answer: Ethidium bromide is a fluorescent dye commonly used to stain DNA and visualise it under ultraviolet (UV) light. When ethidium bromide binds to DNA, it intercalates between the base pairs of the DNA molecule.

When exposed to UV radiation, ethidium bromide emits a fluorescent signal that can be visualised. Under UV light, DNA stained with ethidium bromide typically appears as bright orange bands or spots.

The orange colour is a result of the interaction between ethidium bromide and the DNA molecule. The intercalation of ethidium bromide into the DNA helix causes a shift in the absorbance and emission wavelengths, resulting in the characteristic orange fluorescence when exposed to UV light.

Hence, the correct answer is option 4) Bright orange colour.

9. In the gene gun method used to introduce alien DNA into host cells, microparticles of _____ metal are used.

Option 1: Copper

Option 2. Zinc

Option 3: Tungsten or gold

Option 4. Silver

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Difficulty Level: Medium

Answer: In the gene gun method, which is a technique used to introduce foreign DNA into host cells, microparticles of either tungsten or gold are commonly used.

The gene gun method, also known as a biolistic particle delivery system, involves coating tiny microparticles with foreign DNA and propelling them into the target cells. The microparticles act as carriers for the foreign DNA and are propelled into the host cells using a high-pressure gas or a mechanical force.

Tungsten and gold are preferred materials for microparticles because they are dense and can effectively penetrate the cell walls and membranes of the target cells. The microparticles are typically coated with the desired DNA and then shot into the host cells, where they can integrate into the cell's genome and express the foreign genes.

Therefore, in the gene gun method, microparticles of either tungsten or gold are commonly used as carriers for the introduction of alien DNA into host cells.

Hence, the correct option is Option 3) Tungsten or gold.

10. The main steps in the formation of recombinant DNA are given below. Arrange these steps in a correct sequence.

A. Insertion of recombinant DNA into the host cell.

B. Cutting of DNA at a specific location by a restriction enzyme.

C. Isolation of desired DNA fragment.

D. Amplification of the gene of interest using PCR.

Choose the correct answer from the options given below:

Option 1: B, C, D, A

Option 2: C, A, B, D

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Option 3: C, B, D, A

Option 4: B, D, A, C

Difficulty Level: Medium

Answer: The correct sequence of steps in the formation of recombinant DNA is:

B, C, D, A

The correct sequence is as follows:

Step 1: Cutting of DNA at specific locations by a restriction enzyme (B)
Step 2: Isolation of the desired DNA fragment (C)
Step 3: Amplification of the gene of interest using PCR (D)
Step 4: Insertion of the recombinant DNA into the host cell (A)

Hence, the correct answer is Option 1—B, C, D, A

11. Which of the following is not a cloning vector?

Option 1. BAC

Option 2. YAC

Option 3. pBR322

Option 4. Probe

Difficulty Level: Medium

Answer: A probe is not a cloning vector but is a short single-stranded DNA or RNA molecule designed to hybridise with a sample to detect specific nucleotide sequences. On the other hand, the cloning vectors are Bacterial Artificial Chromosomes (BAC), Yeast Artificial Chromosomes (YAC), and plasmid vectors such as pBR322, used for the replication and cloning of DNA fragments. Such vectors allow for the introduction, storage, and transfer of foreign DNA into host organisms and are indispensable to molecular biology and genetic engineering.

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Hence, the correct answer is Option (4) Probe.

NEET 2022

Chapter: Biotechnology and Its Applications

12. Transposons can be used during which one of the following?

Option 1: Gene silencing

Option 2. Autoradiography

Option 3. Gene sequencing

Option 4. Polymerase Chain Reaction

Difficulty Level: Difficult

Answer: **Option 1** is correct because the source of complementary RNA for RNAi (RNA interference) or gene silencing could be transposons (mobile genetic elements).

Option 2 is incorrect, as autoradiography usually follows hybridisation

Option 3 is incorrect, as transposons are not required in gene sequencing.

Option 4 is also incorrect, as PCR (polymerase chain reaction) is a process to make copies of DNA and does not need transposons.

Hence, the correct answer is option (1) Gene silencing

13. DNA polymorphism forms the basis of:

Option 1: DNA fingerprinting

Option 2. Both genetic mapping and DNA finger-printing

Option 3. Translation

Option 4. Genetic mapping

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Difficulty Level: Medium

Answer: DNA polymorphism is a difference in DNA sequence among individuals, groups, or populations.

Genetic mapping refers to the process of determining the location of genes on chromosomes.

DNA fingerprinting is a laboratory technique used to determine the probable identity of a person based on the nucleotide sequences of certain regions of human DNA that are unique to individuals.

Therefore, DNA polymorphism forms the basis of genetic mapping and DNA fingerprinting.

Hence, the correct option is (2) Both genetic mapping and DNA finger-printing

14. Statements related to human insulin are given below:

Which statement(s) is/are correct about genetically engineered insulin?

(a) Pro-hormone insulin contains an extra stretch of C-peptide

(b) A-peptide and B-peptide chains of insulin were produced separately in E.coli, extracted and combined by creating a disulphide bond between them

(c) Insulin used for treating Diabetes was extracted from Cattles and Pigs

(d) Pro-hormone Insulin needs to be processed for conversion into a mature and functional hormone

(e) Some patients develop allergic reactions to the foreign insulin.

Choose the most appropriate answer from the options given below:

Option 1. (b) only

Option 2. (c) and (d) only

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Option 3: (c), (d), and (e) only

Option 4. (a), (b), and (d) only

Difficulty Level: Difficult

Answer: Let's evaluate the given statements concerning genetically engineered insulin:

(a) Correct. Pro-hormone insulin is characterised by the presence of a C-peptide segment. Upon processing, this C-peptide is eliminated to yield functional mature insulin.

(b) Accurate. In the synthesis of human insulin with E. coli, the A- and B-peptide chains are generated independently. These chains are then joined to create the functional hormone structure via disulphide bond formation.

(c) Inaccurate. Initially, insulin for diabetes treatment was isolated from bovine and porcine pancreases. The introduction of recombinant DNA technology marked a significant shift in its production.

(d) Correct. Pro-hormone insulin undergoes processing to excise the C-peptide and convert it into the active hormonal form.

(e) Incorrect. Allergic responses can occur in patients using insulin from non-human sources, such as cattle or pigs. However, these reactions are less common with recombinant insulins.

Hence, the correct answer is option 4) (a), (b), and (d) only.

15. Given below are two statements: one is labelled as Assertion (A), and the other is labelled as Reason (R).

Assertion (A): Polymerase chain reaction is used in DNA amplification

Reason (R): The ampicillin-resistance gene is used as a selectable marker to check transformation In light of the above statements, choose the correct answer from the options given below :

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Option 1. Both (A) and (R) are correct, but (R) is not the correct explanation of (A) below:

Option 2. (A) is correct, but (R) is not correct

Option 3. (A) is not correct, but (R) is correct

Option 4. Both (A) and (R) are correct, and (R) is the correct explanation of (A)

Difficulty Level: Medium

Answer: →PCR stands for Polymerase Chain Reaction. In this reaction, multiple copies of the gene (or DNA) of interest are synthesised in vitro using two sets of primers and the enzyme DNA polymerase.

In this reaction, each cycle has three steps:

i) Denaturation

ii) Primer annealing

iii) Extension of primers.

→ Selectable markers help in identifying and eliminating non-transformants and selectively permitting the Growth of the transformants.

Transformation is a procedure through which a piece of DNA is introduced into a host bacterium.

The genes encoding resistance to antibiotics are Ampicillin, chloramphenicol, tetracycline, Kanamycin, etc.

Hence, the correct answer is Option 1. Both (A) and (R) are correct, but (R) is not the correct explanation of (A)

16. In gene therapy for Adenosine Deaminase (ADA) deficiency, the patient requires periodic infusion of genetically engineered lymphocytes because

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Option 1. Gene isolated from marrow cells producing ADA is introduced into cells at embryonic stages.

Option 2. Lymphocytes from the patient's blood are grown in culture outside the body

Option 3. Genetically engineered lymphocytes are not immortal cells.

Option 4. A retroviral vector is introduced into these lymphocytes.

Difficulty Level: Medium

Answer: In gene therapy for Adenosine Deaminase (ADA) deficiency, the patient requires periodic infusion of genetically engineered lymphocytes because these modified lymphocytes have a limited lifespan in the body. Therefore, to maintain therapeutic ADA levels, repeated infusions of these engineered cells are necessary.

Hence, the correct answer is Option (2): Lymphocytes from the patient's blood are grown in a culture outside the body.

NEET 2021

Chapter: Biotechnology: Principles & Processes

17. Given below are two statements:

Statement I:

Restriction endonucleases recognise specific sequences to cut DNA known as palindromic nucleotide sequences.

Statement II:

Restriction endonucleases cut the DNA strand a little away from the centre of the palindromic site.

In light of the above statements, choose the most appropriate answer from the options given below:

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Option 1: Both Statement I and Statement II are incorrect

Option 2. Statement I is correct, but Statement II is incorrect

Option 3. Statement I is incorrect, but Statement II is correct

Option 4. Both Statement I and Statement II are correct

Difficulty Level: Medium

Answer: Statement 1: Restriction Endonucleases are 'molecular scissors.'. These recognise special sequences known as palindromic sequences.
This is true. Restriction endonucleases are colloquially referred to as "molecular scissors." They cleave DNA, cutting it into pieces at known sites. Endonucleases recognise and hybridise with palindromic sequences. A palindromic sequence reads the same forward and reverse on the complementary strand of DNA. An example of such a sequence would be "GAATTC"; its complementary strand reads "GAATTC."
Statement 2: These enzymes cut the strand a little away from the centre.
This is wrong concerning how most of the restriction enzymes work. Many restriction endonucleases, indeed, cut within specific distances to the recognition sequence (as would be considered "offset" cuts); most of the most common restriction enzymes cut centrally in the palindrome, creating either blunt ends or overhanging ones, sticky in most cases. The phrase "a little away from the centre" might imply that they cut outside their recognition sites, which is not correct for most of the well-known restriction enzymes.

Hence, the correct answer is Option 4. Both Statement I and Statement II are correct

18. In the following palindromic base sequences of DNA, which one can be cut easily by a particular restriction enzyme?

Option 1: 5′ GAATTC3′; 3′ CT T A A G 5′

Option 2. 5′CTCAGT3′; 3′ GAGTCA 5′

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Option 3. 5′ G T A T T C3′ ; 3′ CATA A G5′

Option 4. 5′ GATACT3′; 3′ CTATGA 5′

Difficulty Level: Medium

Answer: Palindromic Sequences

Palindromic sequences refer to nucleic acid sequences that read the same when their strands are reversed. It is an important feature in many biological processes, including DNA replication and recognition by restriction enzymes.

How To Identify Palindromic Sequences

In light of your question, only option one is said to contain a palindromic sequence, while the others do not. To check this, one would have to look closely at each sequence and determine whether or not it reads the same both ways, top and bottom or left and right.

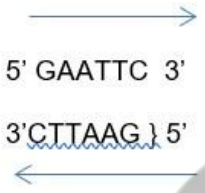
Example

One of the most famous examples of a DNA palindromic sequence is:

5′ - GAATTC - 3′

3′ - CTTAAG - 5′

Both strands read the same as the same strand in the 5′ to 3′ direction.



Hence, the correct answer is Option 1: 5′ GAATTC3′; 3′ CT T A A G 5′

19. Which one of the following statements is not true regarding the gel electrophoresis technique?

Option 1. The separated DNA fragments are stained by using ethidium bromide.

Option 2. The presence of chromogenic substrate gives blue-coloured DNA bands on the gel.

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Option 3. Bright orange-coloured bands of DNA can be observed in the gel when exposed to UV light.

Option 4. The process of extraction of separated DNA strands from gel is called elution.

Difficulty Level: Medium

Answer: In agarose gel electrophoresis, bands are visualised as orange bands with EtBr (Ethiclium Bromide). EtBr on intercalation with nitrogenous bases gives orange-coloured bands on the gel, and it shows fluorescence upon UV exposure. Protein bands stain blue on SDS-PAGE.

Hence, the correct answer is option 2) The presence of chromogenic substrate gives blue-coloured DNA bands on the gel.

20. Which of the following is not a desirable feature of a cloning vector?

Option 1: Presence of a maker gene

Option 2: Presence of single restriction enzyme site

Option 3. Presence of two or more recognition sites

Option 4. Presence of origin of replication

Difficulty Level: Medium

Answer: An ideal cloning vector would contain a recognition site for every restriction enzyme singly so that fragmentation of the vector by the presence of multiple sites for the same enzyme does not occur, minimising the overall efficiency of cloning. The design above permits the efficient insertion of foreign DNA at the precise location desired without unwanted cleavage, permitting successful ligation and propagation in host cells of the recombinant DNA to successfully carry out a cloning experiment.

Hence, the correct answer is Option (3), the presence of two or more recognition sites.

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Chapter: Biotechnology: Principles & Processes

21. Plasmid pBR322 has a PstI restriction enzyme site within gene amp^R that confers ampicillin resistance. If this enzyme is used for inserting a gene—galactoside production—and the recombinant plasmid is inserted in an E. coli strain.

Option 1. It will not be able to confer ampicillin resistance to the host cell.

Option 2. The transformed cells will have the ability to resist ampicillin as well as produce galactoside.

Option 3. It will lead to lysis of the host cell.

Option 4. It will be able to produce a novel protein with a dual ability.

Difficulty Level: Medium

Answer: A gene called ampR found on the plasmid pBR322 renders E. coli resistant to the antibiotic ampicillin. This gene contains a unique location that can be cut by the restriction enzyme PstI.
The ampR gene will be broken if a new gene, such as the gene for galactoside synthesis, is inserted using PstI. At that point, the plasmid can no longer provide ampicillin resistance to the cell.
E. Coli carrying this recombinant plasmid will,l therefore, not be able to thrive on plates containing ampicillin. Scientists can determine whether or not the plasmid has the new gene in this way.

Hence, the correct answer is option 1)It will not be able to confer ampicillin resistance to the host cell.

22. DNA fingerprinting involves identifying differences in some specific regions in DNA sequence, called

Option 1. Satellite DNA

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Option 2. Repetitive DNA

Option 3. Single nucleotides

Option 4. Polymorphic DNA

Difficulty Level: Medium

Answer: **Repetitive DNA:** Stretches of DNA that repeat themselves throughout a genome, either in tandem or interspersed along the genome. These stretches can comprise up to fifty per cent or more of an organism's DNA.

Hence, the correct answer is option (2) Repetitive DNA

23. A specific recognition sequence identified by endonucleases to make cuts at specific positions within the DNA is:

Option 1. Degenerate primer sequence

Option 2. Okazaki sequences

Option 3. Palindromic Nucleotide sequences

Option 4. Poly (A) tail sequences

Difficulty Level: Medium

Answer: A specific part of DNA that endonucleases can identify and cut is termed a restriction site or recognition sequence. These are like molecular scissors, with each enzyme searching for a distinct nucleotide pattern, typically 4-8 bases long. An example is EcoRI, which finds the sequence 5'-GAATTC-3' and cuts just after the first G. Another is HindIII, which locates 5'-AAGCTT-3' and snips between the two A's. These enzymes are vital in fields like genetic engineering and molecular biology for creating gene copies, mapping DNA, and other useful lab techniques. They help us to fragment DNA into smaller segments with either sticky or blunt ends, which can be rearranged as necessary for different experiments.

Hence, the correct answer is option 3) Palindromic Nucleotide sequences.

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24. DNA strands on a gel stained with ethidium bromide, when viewed under UV radiation, appear as:

Option 1. Yellow bands

Option 2: Bright orange bands

Option 3. Dark red bands

Option 4. Bright blue bands

Difficulty Level: Medium

Answer: DNA strands on a gel stained with ethidium bromide, when viewed under UV radiation, appear as bright orange bands as EtBr absorbs UV radiation at 210-285 nm wavelength and gets excited at 605 nm, due to which its fluorescent orange colour.

Hence, the correct option is (2) bright orange bands.

25. Which of the following is not an application of PCR (Polymerase Chain Reaction)?

Option 1: Molecular diagnosis

Option 2. Gene amplification

Option 3. Purification of isolated protein

Option 4. Detection of gene mutation

Difficulty Level: Medium

Answer: Some of the areas of application of PCR are mentioned here:

(i) Diagnosis of pathogens.

(ii) Diagnosis of specific mutation.

(iii) DNA fingerprinting.

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(iv) Detection of specific microorganisms.

(v) In prenatal diagnosis.

(vi) Diagnosis of plant pathogens.

(vii) In palaeontology.

Hence, the correct answer is option 3—purification of isolated protein is not an application of PCR (polymerase chain Reaction).

26. During the purification process for recombinant DNA technology, the addition of chilled ethanol precipitates out:

Option 1: RNA

Option 2. DNA

Option 3. Histones

Option 4. Polysaccharides

Difficulty Level: Easy

Answer: The purified DNA, after treatment with various enzymes, precipitates out after the addition of chilled ethanol. The process is known as DNA spooling. This process involves the careful winding of precipitated DNA around a glass rod or a comparable instrument, resulting in a structure that resembles a thread. This method is frequently used in molecular biology laboratories to isolate and gather DNA following extraction, facilitating both the visualisation and manipulation of the genetic material.

Hence, the correct answer is option 2) DNA.

27. Which of the following is the correct sequence of steps in PCR (Polymerase Chain Reaction)?

Option 1: Denaturation, Annealing, Extension

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Option 2. Denaturation, Extension, Annealing

Option 3. Extension, Denaturation, Annealing

Option 4. Annealing, Denaturation, Extension

Difficulty Level: Medium

Answer: **Stages in PCR (Polymerase Chain Reaction):**

1. Denaturation:
Heating to 95°C for 1 minute separates the DNA strands due to breaking hydrogen bonds.

2. Renaturation/Annealing:
Cooling to 55°C allows short DNA molecules, called primers, to attach to specific areas of the DNA target.

3. Extension:
DNA synthesis begins at the 3′ end of the primer. Bases are added to match the DNA template strands.

Hence, the correct answer is option 1) Denaturation, Annealing, Extension.

NEET 2019

Chapter: Biotechnology: Principles & Processes

28. Which of the following equipment is essentially required for growing microbes on a large scale for industrial production of enzymes?

Option 1: BOD incubator

Option 2. Sludge digester

Option 3. Industrial oven

Option 4. Bioreactor

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Difficulty Level: Medium

Answer: Bioreactors are stirrer cylindrical tanks. The stirrer provides movement, even mixing, and availability throughout the bioreaction. Bioreactors can be 100 to 1000 for large production. To produce large quantities of enzymes, bioreactors are used wherein large volumes of culture can be processed or provide optimal conditions for achieving the desired product. Bioreactors maintain controlled conditions such as temperature, pH, and nutrient supply to maximise microbial or cell growth. They can be operated in batch, fed-batch, or continuous modes depending on the production requirements. Specialised bioreactors, such as airlift and fluidised bed reactors, enhance oxygen transfer and mixing efficiency. These systems are widely used in pharmaceuticals, food industries, and biofuel production for large-scale bioprocessing.

Hence, the correct answer is option 4—Bioreactor.

29. The following statements describe the characteristics of the enzyme Restriction Endonuclease. Identify the incorrect statement.

Option 1. The enzyme cuts DNA molecules at the identified position within the DNA.

Option 2. The enzyme binds DNA at specific sites and cuts only one of the two strands.

Option 3. The enzyme cuts the sugar-phosphate backbone at specific sites on each strand.

Option 4. The enzyme recognises a specific palindromic nucleotide sequence in the DNA.

Difficulty Level: Easy

Answer: Restriction enzymes are recognised as restriction endonucleases, and they are special enzymes that cut DNA molecules on or near specific base sequences known as recognition sites.

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- There are three types: Type I, Type II, and Type III, & this division is based on structural and functional criteria. Type I enzymes cut DNA at random sites that are far removed from the recognition sequences, and they require ATP for their action.
- Type II enzymes, which are the most used in molecular biology, cut DNA strands at exact positions within or close to the recognition sequence without being dependent on ATP.
- Type III, like Type I, also requires ATP but cuts DNA a bit further away from the recognition sites and has a different mechanism.

Hence, the correct option is 2) The enzyme binds DNA at specific sites and cuts only one of the two strands.

30. Expressed Sequence Tags (ESTs) refer to:

Option 1. Genes expressed as RNA

Option 2. Polypeptide expression

Option 3. DNA polymorphism

Option 4. Novel DNA sequence

Difficulty Level: Medium

Answer: ESTs are small DNA sequences derived from the expressed RNA, thus expressing genes that are being transcribed. They have emerged as useful tools in gene identification and cataloguing for a genome. Through analysis of ESTs, scientists get insights into gene expression, discover new genes, and characterise alternative splicing events to contribute significantly to genomic research and functional genomics.

Hence, the correct answer is option 1: genes expressed as RNA

31. What triggers the activation of protoxins to activate the Bt toxin of Bacillus thuringiensis in bollworms?

Option 1. Body temperature

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Option 2. The moist surface of midnight

Option 3. Alkaline pH of gut

Option 4. Acidic pH of the stomach

Difficulty Level: Medium

Answer: Bacillus thuringiensis is a Gram-positive bacteria. During sporulation, this bacteria forms crystals of proteins (cry proteins), which are coded by cry genes (Cry 1 AC) Cry genes are used to make insecticide-transgenic plants.

When insects ingest crystal proteins, the alkaline pH of the gut denatures the insoluble proteins into soluble ones. Soluble proteins make the gut porous and cause the death of insects (Lepidopteran). The toxins present in Bacillus thuringiensis remain inactive protoxins, which, when they enter the gut of the insect, get activated. The alkaline pH of the insect gut activates the protoxin into an active toxin.

Hence, the correct answer is option 3) Alkaline pH of the gut

32. Which of the following is true for Golden Rice?

Option 1. It is a vitamin A-enriched product with a gene from a daffodil.

Option 2. It is pest-resistant, with a gene from Bacillus thuringiensis

Option 3. It is drought tolerant, developed using Agrobacterium vector

Option 4. It has a yellow grain because of a gene introduced from a primitive variety of rice.

Difficulty Level: Medium

Answer: Vitamin A-enriched rice was produced to increase nutrient quantity. Golden rice is not commercially available. Golden rice was developed by IRRI to address vitamin A deficiency, which is a serious public health problem affecting millions of children. Golden rice was genetically engineered to produce high levels of beta-carotene biosynthesis genes:

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1. psy (phytoene synthase) from daffodil

2. cruel (phytoene desaturase) from soil bacterium Erwinia uredovera

The psy and coral genera were transferred into the rice genome under the endosperm-specific promoter so that they are only expressed in the endosperm.

Thus, golden rice is a vitamin A-enriched rice with a gene from daffodil.

Hence, the correct answer is option 1, which is a vitamin A-enriched one with a gene from daffodil.

NEET 2018

Chapter: Biotechnology: Principles & Processes

33. The correct order of steps in Polymerase Chain Reaction (PCR) is

Option 1: Denaturation, Annealing

Option 2. Annealing, Extension, Denaturation

Option 3. Extension, Denaturation, Annealing

Option 4. Denaturation, Annealing, Extension

Difficulty Level: Medium

Answer: As we learnt in

PCR -

The polymerase chain reaction (PCR) was invented by Kary Mullis (1985). In PCR, Template DNA, Oligonucleotides or primers, Taq polymerase and dNTP are used to amplify DNA

- wherein

PCR is used to amplify (many copies) targeted sequence

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Polymerase Chain Reaction (PCR) is used for making multiple copies of the gene (or DNA) of interest in vitro. Each cycle has three steps

(i) Denaturation

(ii) Primer annealing

(iii) Extension of primer

Hence, the correct answer is Option Denaturation, Annealing, Extension

34. The use of bioresources by multinational companies and organisations without authorisation from the concerned country and its people is called

Option 1. Biodegradation

Option 2. Biopiracy

Option 3. Bio-infringement

Option 4. Bioexploitation

Difficulty Level: Medium

Answer: Biopiracy—The use of bioresources by multinational companies and other organisations without proper authorisation from the countries and people concerned and compensatory payment. Biopiracy is a term used for or refers to the use of bioresources by multinational companies and other organisations without proper authorisation from the countries and people concerned with compensatory payment.

Hence, the correct answer is option 2) Biopiracy

NEET 2018

Chapter: Biotechnology and Its Applications

35. In India, the organisation responsible for assessing the safety of introducing genetically modified organisms for public use is

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Option 1. Research Committee on Genetic Manipulation (RCGM)

Option 2. Council for Scientific and Industrial Research (CSIR)

Option 3. Indian Council of Medical Research (ICMR)

Option 4. Genetic Engineering Appraisal Committee (GEAC)

Difficulty Level: Medium

Answer: Genetically modified crops (GMOs)—These are transgenic crops that contain and express one or more useful foreign genes or transgenes. The food that is prepared by GMC is called genetically modified food.

Ex. BT Cotton, Flavr Savr Tomato.

Transgenic plants, also known as bioreactors or factories, produce specific products.

The Indian government has set up organisations such as GEAC (Genetic Engineering Appraisal Committee), which will make decisions regarding the validity of GM research and the safety of introducing GM organisms for public services.

Hence, the correct answer is option 4, Genetic Engineering Appraisal Committee (GEAC)

36. Which of the following is commonly used as a vector for introducing a DNA fragment in human lymphocytes?

Option 1: λ phage

Option 2. Ti plasmid

Option 3. Retrovirus

Option 4. pBR 322

Difficulty Level: Difficult

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Answer: Gene Therapy: If a person is diagnosed with a defective gene leading to disability, they may get this gene replaced by a normal function or may like to get this gene repaired, called gene therapy.

The 1st gene therapy in animals was carried out in mice to correct the dwarfness by introducing a growth hormone gene.

Lymphocytes from the patient's blood are grown in a culture outside the body. A functional gene is introduced into these lymphocytes using a retroviral vector. Retrovirus is commonly used as a vector for introducing a DNA fragment into human lymphocytes.

Hence, the correct answer is option 3) Retrovirus.

NEET 2017

Chapter: Biotechnology and Its Applications

37. Which of the following steps is catalysed by Taq DNA polymerase in a PCR reaction?

Option 1. Denaturation of template DNA

Option 2. Annealing of primers to template DNA

Option 3. Extension of primer end on the template DNA

Option 4. All of the above

Difficulty Level: Difficult

Answer: The extension of the primer end on the template DNA occurs during DNA replication or PCR amplification. Once the primer binds to the template strand, DNA polymerase extends the primer by adding complementary nucleotides to the growing strand. This process occurs in the 5′ to 3′ direction, with the polymerase synthesising the new strand based on the template sequence. The extension continues until the entire segment of DNA has been replicated or amplified.

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Hence, The correct answer is option (3) Extension of the primer end on the template DNA

38. In RNAi, genes are silenced using:

Option 1: ss DNA

Option 2. ds DNA

Option 3. ds RNA

Option 4: ss RNA

Difficulty Level: Medium

Answer: In RNA interference (RNAi), genes are silenced using small pieces of RNA. These tiny strands, called small interfering RNA (siRNA), match up with the messenger RNA (mRNA) from a gene. When they connect, the mRNA gets broken down and can no longer make proteins. This process is like turning off a light switch; when the gene is silenced, it stops working. Scientists use this method to study genes and understand how they affect health and diseases. By controlling which genes are active, researchers can learn how to develop new treatments and therapies. RNAi shows great promise in medicine by helping to target specific genes involved in illnesses.

Hence, the correct answer is option 3) ds RNA

39. In an experiment, bacteria were infected with ^{32}P labelled virus in a ratio of $5:1$. The culture was rigorously shaken, followed by centrifugation. Radioactivity was

Option 1: lost due to metabolic activity.

Option 2: detected in supernatant as inorganic phosphate.

Option 3. detected in the supernatant in association with viral capsid.

Option 4: detected in bacterial cell pellet.

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Difficulty Level: Difficult

Answer: The radioactive phosphorous-labelled DNA remained with the bacterial pellet because the genetic information of the virus was incorporated into the genetic material of the host cell.

After centrifugation, they found that most of the radioactive phosphorus was detected in the cells rather than the supernatant, meaning that the phage DNA must have entered the cells when the phages infected the bacteria. So, radioactivity was found in the cell pellet.

Hence, the correct answer is option (d) detected in the bacterial cell pellet.

40. In biotechnology applications, a selectable marker is incorporated in a plasmid.

Option 1: to increase its copy number

Option 2: to increase the transformation efficiency

Option 3: to eliminate the non-transformants

Option 4. to increase the expression of the gene of interest

Difficulty Level: Difficult

Answer: A selectable marker is a gene introduced into a cell, especially a bacterium or cells in culture, that confers traits suitable for artificial selection, i.e., it allows the selection of positively transformed cells and eliminates the cells.

Hence, the correct answer is an option (3) to eliminate the non-transformants

41. The process of RNA interference has been used in the development of plants resistant to

Option 1: nematodes

Option 2: fungi

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Option 3: viruses

Option 4: insects

Difficulty Level: Easy

Answer: Antisense RNA—It is a single-stranded RNA that is complementary to a protein-coding messenger RNA (mRNA) that hybridises with it and blocks protein translation. This method is also known as gene knockdown. RNA I (Interference) RNA is also used for the same purpose. RNAi is used to block protein synthesis and the development of plants resistant to nematodes.

Hence, the correct answer is option 1—nematodes.

42. Which one of the following is used as a vector for cloning genes into higher organisms?

Option 1. Baculoviruses

Option 2. Salmonella typhimurium

Option 3. Rhizopus nigricans

Option 4. Retrovirus

Difficulty Level: Easy

Answer: Vectors -

The vectors are DNA molecules that can carry a foreign DNA segment and replicate inside the host cell. Vectors may be plasmids, bacteriophages, cosmids, phagemids, yeast artificial chromosomes (YAC), bacterial artificial chromosomes (BAC), transposons, shuttle vectors, and viruses. Vectors carry inserted DNA from one host to another host. Vectors used for cloning genes into higher organisms are retroviruses.

Hence, the correct answer is option 4—Retrovirus

43. Transgenic plants are the ones:

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Option 1. is generated by introducing foreign DNA into a cell and regenerating a plant from that cell.

Option 2: produced after protoplast fusion in an artificial medium.

Option 3: grown in artificial medium after hybridisation in the field.

Option 4. produced by a somatic embryo in an artificial medium.

Difficulty Level: Difficult

Answer: Transgenic plants are the ones in which the desired gene of interest is inserted and produced. Transgenic plants are genetically modified organisms (GMOs) that carry a foreign gene introduced through genetic engineering to express specific traits. These traits may include pest resistance, herbicide tolerance, improved nutritional value, or enhanced shelf life. For example, Bt cotton contains a gene from Bacillus thuringiensis to resist bollworms. Transgenic plants are widely used in agriculture to increase productivity and reduce dependency on chemical pesticides. They also serve as platforms for producing pharmaceutical compounds, such as vaccines and therapeutic proteins.

Hence, the correct answer is option 1—generated by introducing foreign DNA into a cell and regenerating a plant from that cell.

44. Which one of the following palindromic base sequences in DNA can be easily cut at about the middle by some particular restriction enzyme?

Option 1. 5′ CACGTA 3′;

3′ CTCAGT 5′

Option 2. 5′ CGTTCG 3′;

3′ ATGGTA 5′

Option 3. 5′ GATATG 3′;

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3′ CTACTA 5′

Option 4. 5′ GAATTC 3′;

3′ CTTAAG 5′

Difficulty Level: Medium

Answer: Palindromic sequences are read the same from back to front.DNA is cut by EcoRI. Palindromic sequences are read the same from back to front, meaning the sequence is symmetrical on both strands. These sequences play a crucial role in molecular biology, especially in the recognition sites for restriction enzymes. One such enzyme, EcoRI, cuts DNA at the palindromic sequence GAATTC, making a sticky end that can be used in recombinant DNA technology. The ability of EcoRI to cut DNA at specific palindromic sequences is fundamental for genetic manipulation and cloning processes.

Hence, the correct answer is option 4 -

5′ GAATTC 3′;

3′ CTTAAG 5′

45. Which one of the following is used as an actor for cloning genes into higher organisms?

Option 1. Retrovirus

Option 2. Baculovirus

Option 3. Salmonella typhimurium

Option 4. Rhizopus nigricans

Difficulty Level: Difficult

Answer: Retroviral vectors are derived from a retrovirus. It is used for the transfer of genes for higher organisms (mammalian). Retroviral vectors are used in gene therapy

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and molecular biology to introduce specific genes into mammalian cells. They are derived from retroviruses, which can reverse transcribe their RNA genome into DNA, allowing integration into the host's genome. This characteristic makes retroviral vectors useful for stable gene expression in target cells. Retroviral vectors can be engineered to carry therapeutic genes, which are then inserted into the DNA of host cells, offering potential treatments for genetic disorders. However, their use is often limited by the risk of insertional mutagenesis, where the inserted gene disrupts normal gene function.

Hence, the correct answer is option 1—Retrovirus.

46. Which one of the following is now being commercially produced by biotechnological procedures?

Option 1: Nicotine

Option 2. Morphine

Option 3. Quinine

Option 4. Insulin

Difficulty Level: Medium

Answer: Human insulin was developed by Eli Lilly in 1982 using recombinant DNA technology from Escherichia coli. Insulin is in demand for many diabetic patients. Thus, insulin is commercially produced. Recombinant DNA technology allowed the insertion of the human insulin gene into Escherichia coli, enabling the bacteria to produce insulin. This method provided a more consistent and safer supply of insulin compared to previous animal-derived sources. The production process involves culturing the modified bacteria in large fermenters, where they produce insulin that is then purified for medical use. The availability of recombinant human insulin revolutionised diabetes treatment, offering patients a more reliable and effective option. Since its development, recombinant insulin has been widely used to manage blood sugar levels in individuals with diabetes.

Hence, the correct answer is option 4) Insulin

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NEET 2017

Chapter: Biotechnology: Principles and Processes

47. The DNA fragments separated on an agarose gel can be visualised after staining with:

Option 1. Bromophenol blue

Option 2. Acetocarmine

Option 3. Aniline blue

Option 4. Ethidium Bromide

Difficulty Level: Easy

Answer: Ethidium Bromide

The separated DNA fragments can be seen only after staining with a compound known as Ethidium Bromide, followed by exposure to UV radiation as bright orange-coloured bands. Ethidium bromide is a fluorescent dye that intercalates with DNA molecules and fluoresces when exposed to UV light (302 nm). EtBr is used to stain DNA fragments.

Hence, the correct answer is Option 4—Ethidium Bromide.

48. A gene whose expression helps to identify transformed cells is known as

Option 1: Selectable marker

Option 2. Vector

Option 3. Plasmid

Option 4. Structural gene

Difficulty Level: Medium

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Answer: In recombinant DNA technology, selectable markers are the specific genes that are used to identify the transformants from the non-transformants after the process of recombination. These genes are used to detect whether the incorporation of a nucleic acid sequence has been successful into an organism's DNA. These marker genes are resistant to various antibiotics and are used in cloning vectors.

Hence, the correct answer is option 1) Selectable markers.

49. The process of separation and purification of expressed protein before marketing is called :

Option 1. Upstream processing

Option 2. Downstream processing

Option 3. Bioprocessing

Option 4. Post-production processing

Difficulty Level: Medium

Answer: Downstream processing is used for the separation and purification of the desired product before marketing. It is mainly used during large-scale production of metabolites. Hence, the correct answer is 'downstream processing.'.

Hence, the correct option is 2 Downstream processing

50. What is the criterion for DNA fragment movement on agarose gel during gel electrophoresis?

Option 1. The larger the fragment size, the farther it moves

Option 2. The smaller the fragment size, the farther it moves

Option 3. Positively charged fragments move to the farther end

Option 4. Negatively charged fragments do not move

Difficulty Level: Medium

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Answer: Gel Electrophoresis -

This technique is used to separate DNA fragments based on charge and size. Agarose is a polysaccharide that is extracted from red algae (gelidium). Gel electrophoresis is used to separate DNA fragments. The smaller the fragments, the further they will move.

Hence, the correct option is (2) The smaller the fragment size, the farther it moves.

NEET 2016

Chapter: Biotechnology: Principles and Processes

51. Which of the following is a restriction endonuclease?

Option 1. Hind II

Option 2. Protease

Option 3. DNase I

Option 4. RNase

Difficulty Level: Easy

Answer: Restriction Endonuclease—These enzymes cleave DNA only within or near the specific base sequence. These sequences are called recognition sites. They are of three types: RE-I and RE-III. The 1st restriction enzyme (Hind II) was used in RDT. Hind II are restriction endonucleases because they cleave DNA at specific sequences within DNA strands.

Hence, the correct option is 1 Hind I.I.

52. Which of the following is not required for any of the techniques of DNA fingerprinting available at present?

Option 1. Polymerase chain reaction

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Option 2. Zinc finger analysis

Option 3. Restriction enzymes

Option 4. DNA-DNhybridisationon

Difficulty Level: Easy

Answer: Steps of DNA fingerprinting -

(a) Extraction of DNA from the sample cells.

(b) Amplification of DNA by PCR

(c) Restriction digestion of PCR Products

(d) Separation on Agarose gel electrophoresis

(e) Southern blotting

(f) Hybridization using labelled VNTR probe

(g) Detection by autoradiography

- wherein

This technique is widely used in forensic science for criminal cases

In DNA fingerprinting, PCR restriction enzymes and hybridisation techniques are used.

Hence, the correct option is (2) Zinc finger analysis

53. The Taq polymerase enzyme is obtained from

Option 1. Thermus aquatics

Option 2. Thiobacillus ferrooxidans

Option 3. Bacillus subtilis

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Option 4. Pseudomonas putida

Difficulty Level: Difficult

Answer: Tag polymerase is obtained from Thermophilus aquaticus. Taq polymerase can function at high temperatures, which is essential for the process of PCR. During the PCR process, the DNA is denatured by heating it to high temperatures, which separates the two strands of the DNA. Taq polymerase is used to synthesise new strands of DNA by adding nucleotides to the primers, working at temperatures that would denature most other enzymes.

Hence, the correct option is (a) Thermus aquaticus

54. Which of the following is not a feature of the plasmids?

Option: Independent replication

Option 2. Circular structure

Option 3. Transferable

Option 4. Single-stranded

Difficulty Level: Easy

Answer: Plasmids—A plasmid is a small, circular standard DNA molecule that has the power of self-replication. They usually exist in bacteria and some lower eukaryotes. They have antibiotic-resistance genes.

- wherein plasmids provide genetic advantages like antibiotic resistance. Plasmids are double-stranded circular structures.

Hence, the correct option is (4) Single-stranded

55. Which of the following restriction enzymes produces blunt ends?

Option 1. Sal I

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Option 2: Eco RV

Option 3. Xho I

Option 4. Hind III

Difficulty Level: Medium

Answer: The restriction site of Eco RV is GAT_ATC, and it produces blunt ends.

$5' - GAT\ ATC\ - 3'$

$3' - CAT\ TAG\ - 5'$

Restriction enzymes are special enzymes that can recognise sites in the DNA sequences and make a cut at these sites. Restriction enzymes cut the DNA in the middle, leaving blunt ends on the DNA, or staggered cuts are made that leave sticky ends with 5 or 3' overhangs.

Hence, the correct answer is 2) 'Eco Rv'

56. Which of the following is not a component of downstream processing?

Option 1: Separation

Option 2. Purification

Option 3. Preservation

Option 4. Expression

Difficulty Level: Medium

Answer: Downstream processing involves the extraction, separation, purification, and preservation of biotechnological products. Downstream processing doesn't involve the expression of proteins.

This process includes several steps:

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Extraction: It involves isolating the desired product from the culture medium. This can be achieved via the process of filtration, centrifugation, etc.

Separation: This step involves separating the desired product from other cellular components or impurities.

Purification: Purification ensures the removal of unwanted contaminants.

Preservation and Formulation: The final product is stabilised and preserved for storage and transport.

Hence, the correct answer is option 4) expression.

57. A foreign DNA and plasmid cut by the same restriction endonuclease can be joined to form a recombinant plasmid.

Option 1: Eco R1

Option 2. Taq Polymerase

Option 3. Polymerase III

Option 4: ligase

Difficulty Level: Medium

Answer: As we learnt—

Ligase -

These enzymes form phosphodiester bonds between adjacent nucleotides and covalently link two fragments of double-stranded DNA. DNA ligase enzyme joins two DNA fragments together via a phosphodiester bond between a 3' hydroxyl at the end of one fragment and a 5' phosphate at the end of another fragment. Foreign DNA and a plasmid having complementary ends can be ligated using DNA ligase.

Eco R1 is a restriction endonuclease that cuts DNA at specific sites.

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Taq polymerase is a heat-resistant DNA polymerase.

DNA polymerase III is a polymerase that duplicates chromosomal DNA in prokaryotes.

Hence, the correct answer is Option (4) Ligase.

58. Stirred-tank bioreactors have been designed for

Option: Purification of product

Option 2. Addition of preservatives to the product

Option 3. Availability of oxygen throughout the process

Option 4. Ensuring anaerobic conditions in the culture vessel

Difficulty Level: Easy

Answer: Bioreactors -

Bioreactors are stirrer cylindrical tanks. The stirrer provides movement, even mixing, and O_2 availability throughout the bioreaction.

- wherein

Bioreactors can be 100 to 1000. It is for large production.

Bioreactors are used for mixing and O_2 availability throughout the bioreaction.

Hence, the correct option is (3) the availability of oxygen throughout the process.

NEET 2016

Chapter: Biotechnology and Its Applications

59. ELISA technique is based on the principle of

Option 1: DNA replication

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Option 2: Antigen and Antibody interaction

Option 3. Pathogen and Antigen interaction

Option 4. Antigen and Protein interaction

Difficulty Level: Difficult

Answer: **ELISA** – It is a plate-based assay technique designed to detect and quantify soluble substances like proteins, peptides, antibodies, and hormones. The principle is based on antigen-antibody interaction. ELISA (Enzyme-Linked Immunosorbent Assay) uses an enzyme-linked antibody or antigen to produce a detectable signal, often colourimetric. It is widely used in diagnostics, such as detecting infections, autoimmune diseases, and allergens. Variants of ELISA include direct, indirect, sandwich, and competitive assays for diverse applications.

Hence, the correct answer is **(2) antigen and antibody interaction.**

60. Which part of the tobacco plant is infected by Meloidogyne incognita?

Option 1: Flower

Option 2. Leaf

Option 3. Stem

Option 4. Root

Difficulty Level: Difficult

Answer: The roots of the tobacco plants are infected. The RNAi (interference RNA) method is used to develop resistance against infection. Antisense RNA is a single-stranded RNA that is complementary to a protein-coding messenger RNA (mRNA) that hybridises with it and blocks protein translation.

Hence, the correct answer is Option (4) Root.

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